Engineering Product Design – Problem Identification Report

SPPU 2024 | Second Year Computer Engineering

Domain

- Education
- Human-Computer Interaction

Target Users

- Undergraduate students
- First-year students
- Educational institutions

Problem Statement

Traditional display systems require physical screens and touch, limiting interactivity and immersion. There is a need for a touch-free, interactive mid-air display system for enhanced user engagement.

Empathize

Students find conventional displays less engaging

Define

How might we create a touch-free interactive display system?

Ideate

- Fog-based projection
- Gesture interaction

Prototype

Fog screen + projector + ultrasonic sensors

Test

User interaction and visibility testing

Proposed Solution

Fog Projector with Ultrasonic Gesture Control

An innovative mid-air projection system that creates interactive displays on fog screens, enabling touch-free gesture-based interaction using ultrasonic sensors. This solution eliminates the need for physical screens while providing an immersive and engaging user experience for educational applications.

Feasibility & Implementation

- Fog machine
- Projector
- Ultrasonic sensors
- Microcontroller

Expected Outcomes

- Improved student engagement through immersive technology
- Innovative learning experience with touch-free interaction

Conclusion

The Fog Projector with Gesture Interaction represents a significant innovation in educational technology, addressing the need for more engaging and immersive learning experiences. By combining fog-based projection with ultrasonic gesture control, this EPD project demonstrates practical application of human-computer interaction principles while meeting the SPPU 2024 curriculum requirements for Second Year Computer Engineering students.